

Day 1 to Day 6: Python Basics

Duration: 2 Hours per Day

Goal:

Understand Python fundamentals including setup, variables, data types, operators, input/output, control flow, loops, functions, and small coding tasks.

Day 1: Python Setup, Hello World, Variables, Data Types

Python Setup

- Install Python: <https://www.python.org/downloads/>
- IDEs: VS Code, PyCharm, Jupyter Notebook

Hello World

```
print("Hello World")
```

Variables

- Store data in memory.
- No declaration needed, type inferred automatically.

```
x = 10  
name = "Harshith"
```

Data Types

- int, float, str, bool

```
a = 5          # int  
b = 3.14       # float  
c = "AI"       # str  
d = True       # bool
```

Day 2: Operators, Input/Output, Strings

Operators

- Arithmetic: + - * / % // **
- Comparison: == != > < >= <=
- Logical: and or not

Input/Output

```
name = input("Enter your name: ")
age = int(input("Enter your age: "))
print(f"Hello {name}, you are {age} years old")
```

Strings

- Indexing, slicing, methods

```
text = "Hello World"
print(text[0])      # H
print(text[0:5])    # Hello
print(text.lower()) # hello world
```

Day 3: Lists, Tuples

Lists

- Mutable, ordered collection

```
my_list = [1, 2, 3, 4]
my_list.append(5)
print(my_list)
```

Tuples

- Immutable, ordered collection

```
my_tuple = (1, 2, 3)
print(my_tuple[0])
```

Day 4: Dictionaries, Sets

Dictionaries

- Key-value pairs

```
my_dict = {"name": "Harshith", "age": 24}
print(my_dict["name"])
my_dict["city"] = "Bangalore"
```

Sets

- Unordered, unique items

```
my_set = {1, 2, 2, 3}
print(my_set) # {1, 2, 3}
```

Day 5: Conditionals, Loops

Conditionals

```
x = 10
if x > 5:
    print("x is greater than 5")
elif x == 5:
    print("x is 5")
else:
    print("x is less than 5")
```

Loops

For loop

```
for i in range(5):
    print(i)
```

While loop

```
count = 0
while count < 5:
    print(count)
    count += 1
```

Day 6: Functions and Small Coding Tasks

Functions

```
def greet(name):
    print(f"Hello {name}")

greet("Harshith")
```

Function with Return

```
def add(a, b):
    return a + b

result = add(5, 3)
print(result) # 8
```

Small Coding Tasks Examples

1. Even/Odd Check

```
num = int(input("Enter a number: "))
if num % 2 == 0:
    print("Even")
else:
    print("Odd")
```

2. Sum of First N Numbers

```
n = int(input("Enter a number: "))
sum_n = sum(range(1, n+1))
print(sum_n)
```

3. Multiplication Table

```
num = int(input("Enter a number: "))
for i in range(1, 11):
    print(f"{num} * {i} = {num*i}")
```

Key Takeaways (Day 1-6)

- Python basics: setup, syntax, print/input
- Data types and variables
- Operators and expressions
- Strings, lists, tuples, dictionaries, sets
- Conditionals and loops
- Functions and return values
- Small practice programs to reinforce concepts

End of Day 1 to Day 6 Notes.